PAPER REPRODUCTION FOR EFFICIENT NEURAL NETWORKS

PUTTING A RESNET ON A DATA DIET

Can we throw away half of CIFAR-10 and lose nothing?

REPRODUCING PAUL, GANGULI & DZIUGAITE, NEURIPS 2021
AMIR NIGMATULLIN · OCTOBER 2026

ABOUT THE PAPER IN 30 SECONDS

1) Train a few networks for just a few epochs. For every training image, measure how wrong the prediction is (EL2N) or how large its loss gradient is (GraNd).

2) Images with low scores are easy and redundant. Removing them, for example half of CIFAR-10, does not hurt test accuracy.

3) The very highest scores belong to mislabeled or atypical images, and the scores show which examples drive learning.



NOT ALL TRAINING EXAMPLES ARE EQUAL

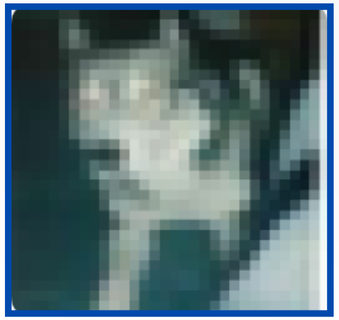


Easy

A typical car. After a few epochs the network classifies it confidently. Showing it again teaches almost nothing.

Hard

These are real CIFAR-10 labels. Atypical or mislabeled images keep producing large errors during training.



“dog”



“dog”



“bird”

WHY FIND THE IMPORTANT EXAMPLES?

1 Compute

Models and datasets keep growing. A smaller training set saves time and energy in every future training run.

2 Small devices

On-device and memory-limited training needs smaller datasets.

3 Choosing data

Online and active learning must decide which examples to keep or to label.

4 Understanding

Which examples shape the decision boundary, and which ones the network barely needs?

EARLIER APPROACH: FORGETTING EVENTS

Toneva et al. (2019) record after every epoch whether each training example is classified correctly. A forgetting event is a switch from correct (✓) to wrong (✗).

Typical car: 0 events	❌	✅	✅	✅	✅	✅	✅	✅	✅	✅
Odd “dog”: 3 events	❌	✅	❌	✅	✅	❌	❌	✅	❌	✅

What it gives

Examples that are never forgotten can be removed without hurting test accuracy.=

The catch

The counts need most of training: on CIFAR-10 the ranking is good after ~25 of 200 epochs and stable after ~75.

Setting and notation

Symbol	Meaning
(x, y)	a training image x and its label y as a one-hot vector, e.g. y = (0, 1, 0, …) for class 2
$f(w, x)$	the network output for weights w: K logits, one per class
$p(w, x) = \text{softmax}(f)$	predicted class probabilities, they sum to 1
$\ell = -\Sigma_k\, y_k \log p_k$	cross-entropy loss of one example
$g_t(x, y) = \nabla \ell(w_t; x, y)$	gradient of the loss of one example with respect to the weights at time t
$w_{t+1} = w_t - \eta\, \Sigma_j\, g_t(x_j, y_j)$	one SGD step: the update is a sum of per-example gradients over the minibatch
w_t is random	it depends on the random initialization and the minibatch order

KEY IDEA:

THE INFLUENCE OF ONE EXAMPLE

How much does training example j change the loss on some other example x* during one SGD step?

1

The step moves the weights by $\Delta w = -\eta \sum_j g_t(x_j, y_j)$, one term per example in the minibatch.

2

To first order the loss on x* changes by $\nabla l(x^*) \cdot \Delta w$; example j adds $-\eta \nabla l(x^*) \cdot g_t(x_j, y_j)$.

3

By Cauchy–Schwarz this contribution is at most $\eta \|\nabla l(x^*)\| \cdot \|g_t(x_j, y_j)\|$.

- The factor $\eta \|\nabla l(x^*)\|$ is the same for every training example j. So the gradient norm $\|g_t(x_j, y_j)\|$ ranks training examples by how much they can influence learning on the rest of the data. This is Lemma 2.2 of the paper.

GRAND: THE EXPECTED GRADIENT NORM

$$\chi_t(x, y) = \mathbb{E} \, \| g_t(x, y) \|_2$$

Why the expectation

The weights w_t are random. One network gives a noisy score, so the paper averages over 10 independent training runs.

How to read it

Low GraNd: the example barely influences learning at time t and can be removed. High GraNd: it strongly shapes the updates.

Caveat

It is an upper bound. Large gradients can cancel each other, so a high score does not guarantee importance.

FROM GRADIENTS TO ERRORS

EL2N

EL2N score = $\mathbb{E}\,\|\,p(w_t, x) - y\,\|_2$, the norm of the error vector. It needs one forward pass and no gradients. The paper finds the approximation accurate after a few epochs, and EL2N turns out to be the stronger pruning score.

1

Chain rule for cross-entropy: $g_t(x, y) = \Sigma_k\,(p_k - y_k)\,\psi_k(x)$, where $\psi_k(x)$ is the gradient of the k-th logit.

2

Hence $\|g\|^2 = \Sigma_k\,\Sigma_l\,(p_k - y_k)(p_l - y_l)\,\langle\psi_k, \psi_l\rangle$.

3

If logit gradients of different classes are nearly orthogonal and of similar size C, only the k = l terms remain: $\|g\| \approx C \cdot \|p - y\|$.

EL2N ON A TOY EXAMPLE: CAT, DOG, BIRD

Easy: a typical cat
label y = (1, 0, 0)
prediction p = (0.90, 0.05, 0.05)
error p – y = (–0.10, 0.05, 0.05)
EL2N = √0.015 ≈ 0.12

Hard: a cat-like “dog”
label y = (0, 1, 0)
prediction p = (0.60, 0.30, 0.10)
error p – y = (0.60, –0.70, 0.10)
EL2N = √0.86 ≈ 0.93



easy cat 0.12

odd dog 0.93

The scale runs from 0 (confident and correct) to √2 ≈ 1.41 (confident and wrong). The hardest CIFAR-10 images in our runs score about 1.41.

HOW THE PAPER USES THE SCORES

1 SCORE

Train 10 networks for a few epochs and average the score of every training example.

2 PRUNE

Remove the fraction of examples with the lowest scores.

3 RETRAIN

Train a new network from a new random seed on the remaining examples and measure test accuracy.

- Setup. ResNet-18 and ResNet-50 on CIFAR-10, CINIC-10 and CIFAR-100. Baselines: the full dataset and a random subset of the same size. Each point is the mean of 4 runs; bands show the 16th–84th percentile.

EXPERIMENT 1: HOW MUCH DATA CAN GO?

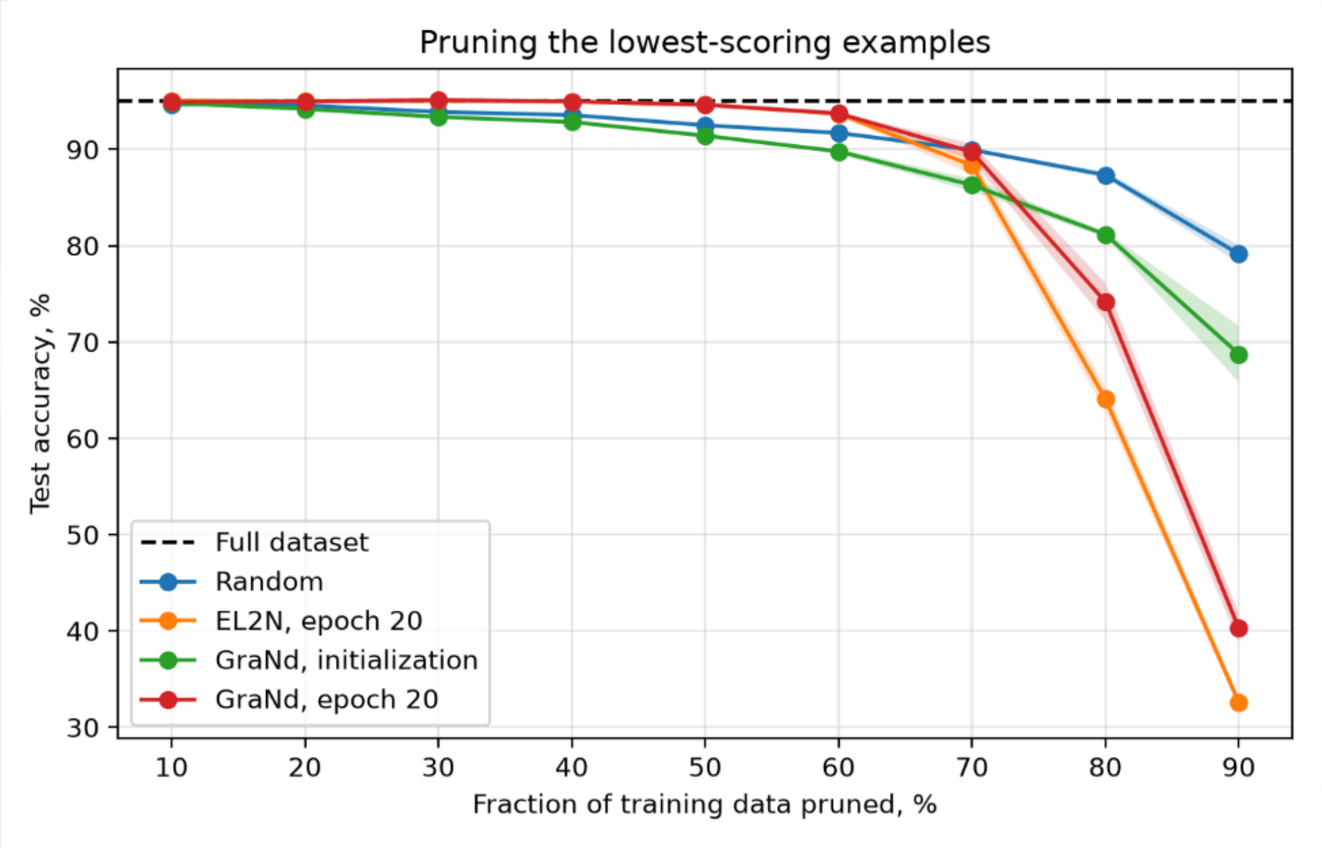


Illustration: our reproduction of Fig. 1 (bottom), ResNet-18, CIFAR-10

What is plotted

Test accuracy (y) after removing a fraction of the lowest-scoring examples (x). Dashed line: all data.

Result

With EL2N from epoch 20, half of CIFAR-10 can be removed while accuracy even rises slightly. On CIFAR-100, 25% can be removed without loss.

Versus forgetting

EL2N from epoch 20 matches forgetting scores that need the whole training run.

WHY REMOVING TOO MUCH BACKFIRES

At extreme pruning levels the paper sees a sharp drop. Keeping only the highest-scoring examples leaves almost no typical images, so whole subpopulations that appear in the test set disappear from training. The authors call this bad coverage of the data distribution.

32.6% test accuracy with only the 10% highest-EL2N images (our runs)

79.1% test accuracy with a random 10% of the data

Studying only trick questions doesn’t teach the basics.

EXPERIMENT 2: HOW EARLY CAN WE SCORE?

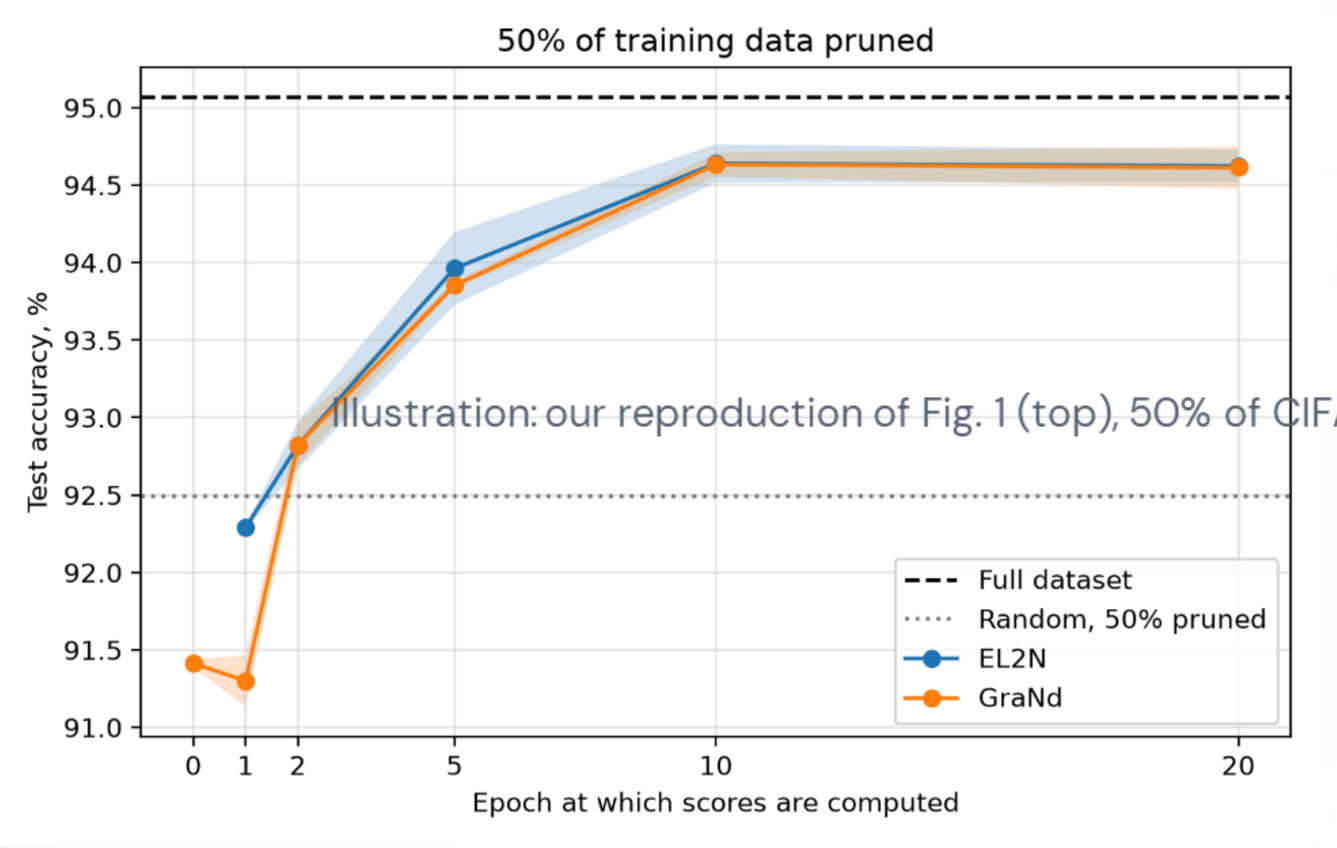


Illustration: our reproduction of Fig. 1 (top), 50% of CIFAR-10 kept

What is plotted

The subset size is fixed. The x-axis is the epoch at which the scores are computed.

Result

EL2N works after only a few epochs. The error early in training already shows which examples will matter for the rest of training.

Versus forgetting

Scores from one network work worse than scores averaged over 10 runs.

ERRATUM: SCORING AT INITIALIZATION DOES NOT WORK

What happened

The NeurIPS 2021 version reported good pruning with GraNd at initialization, before any training. Kirsch (2023) traced this to a bug in the Flax library. The corrected paper says scores at initialization do not perform well.

Why it fails

An untrained network has no idea about the classes. In our runs its gradient norm mostly reflects the pixel norm of the image (rank correlation 0.855), and pruning by it is worse than random.

*Some training is required: a few epochs are enough, zero is not.

A PROPERTY OF THE DATA, NOT OF THE NETWORK

1 DEEPER NETS

ResNet-18 and ResNet-50 on CIFAR-10 give similar curves: the same amount of data can be removed.

2 SCORES TRANSFER

EL2N computed with one architecture or set of hyperparameters prunes data well for another.

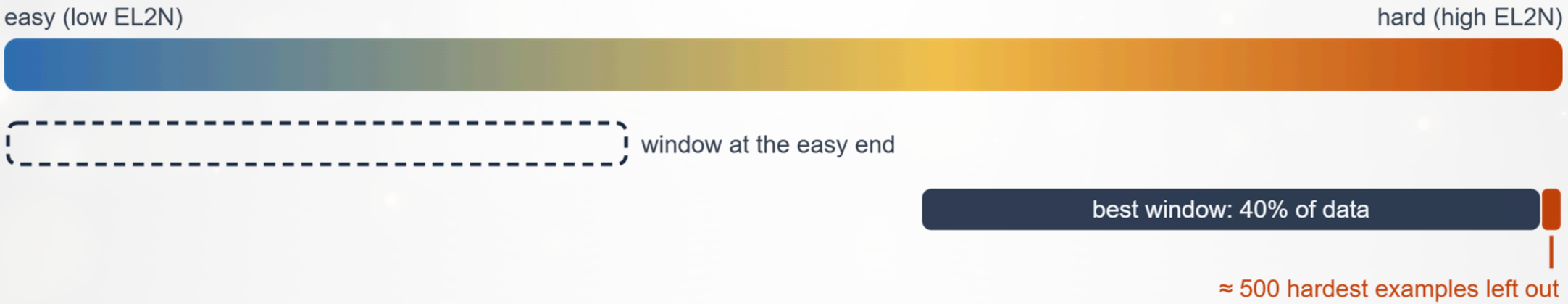
3 AVERAGING

Averaging over runs removes the dependence on specific weights and keeps what belongs to the dataset.

- So the scores can be computed once and the pruned dataset reused for many future trainings.

EXPERIMENT 3: ARE THE HARDEST THE BEST?

The paper sorts all examples by EL2N (from epoch 10) and keeps a window of 40% of the data, sliding it from the easy end towards the hard end.



If the hardest examples were the most valuable, accuracy would keep rising until the window touches the hard end. It does not.

RESULT: SKIP THE VERY HARDEST EXAMPLES

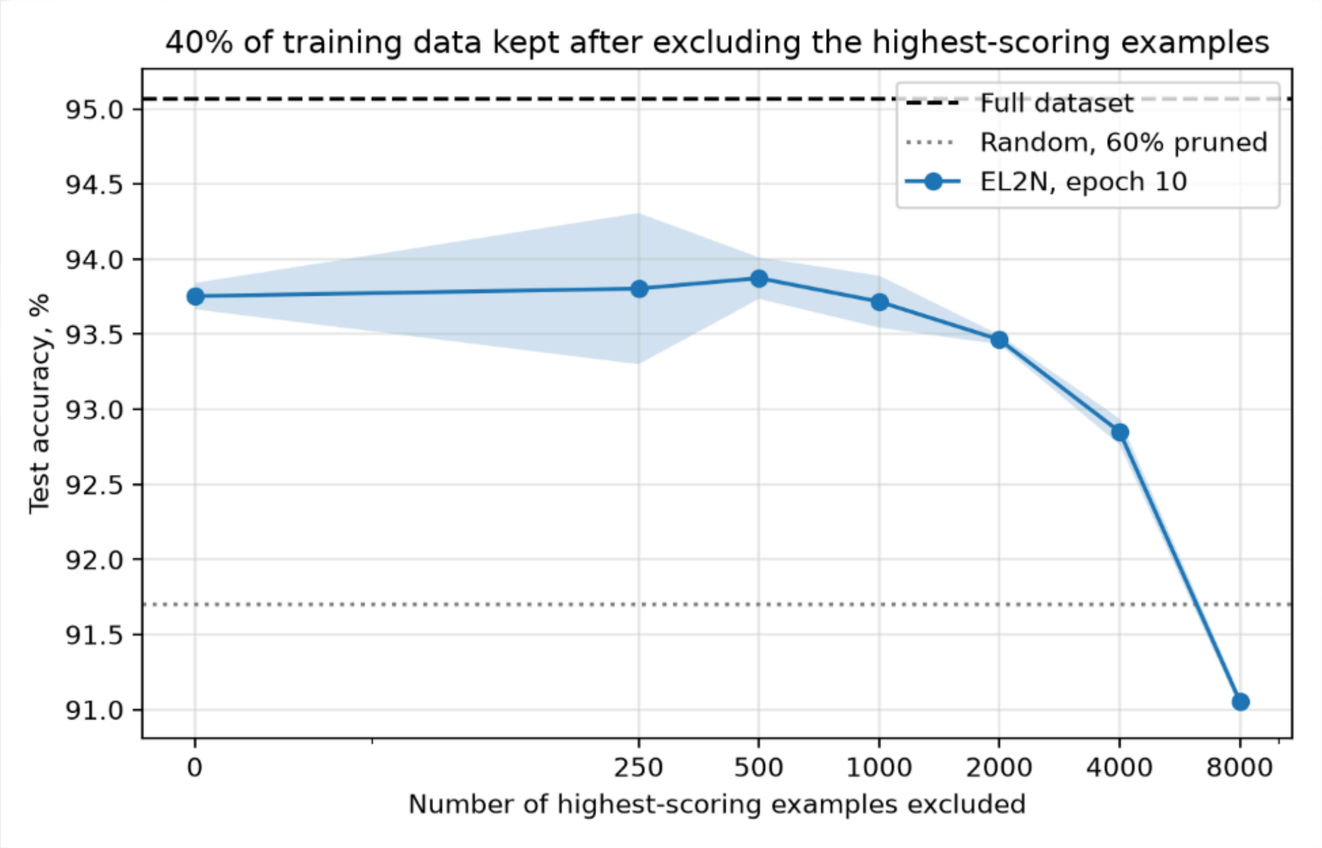


Illustration: our reproduction of Fig. 2 (left), clean labels

Clean labels

Accuracy grows as the window moves to harder examples, but the best window leaves out about 500 of the hardest ones.

With label noise

When 10% of labels are randomized, the best window moves further and leaves out more examples.

In practice

Window start and width are two hyperparameters to tune on a validation set.

WHAT DO THE HIGHEST SCORES LOOK LIKE?

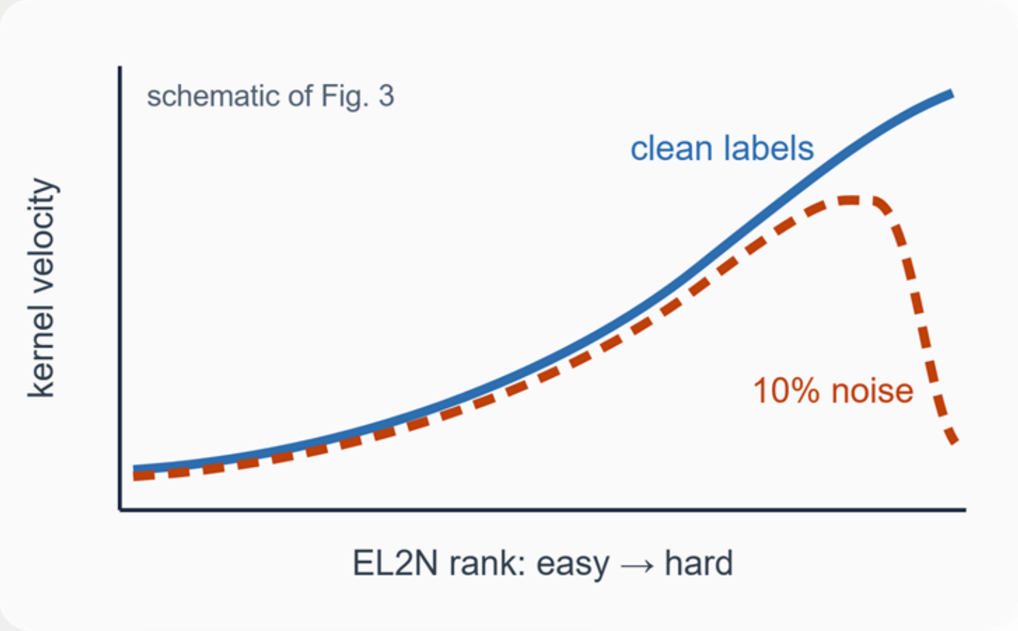


Lowest EL2N (top row). Typical, centered cars and ships. The network is sure about them.

Highest EL2N (bottom row). Outliers of a class, unusual backgrounds or angles, and label noise: several “dogs” look like cats or a horse.

*From our reproduction: EL2N at epoch 20, CIFAR-10.

INSIDE TRAINING: WHO SHAPES THE FEATURES?



1

The NTK in one sentence

For two inputs, the neural tangent kernel measures how similarly the network outputs react to a small change of the weights. Over many inputs it describes the features the network currently uses.

2

What is measured

Kernel velocity: how much this kernel changes from epoch t to t+1 (cosine distance), on blocks of 100 examples with neighboring EL2N.

3

Result

Higher EL2N means faster change at every epoch: high-score examples drive feature learning. With 10% label noise the velocity drops for the very top scores: the network gives up on them.

INSIDE TRAINING: WHEN IS IT DECIDED?

The test

Train to time t*, copy the network, and continue both copies with different minibatch orders. At the end, walk along the straight line between the two final weights and record the error.

1

Error barrier

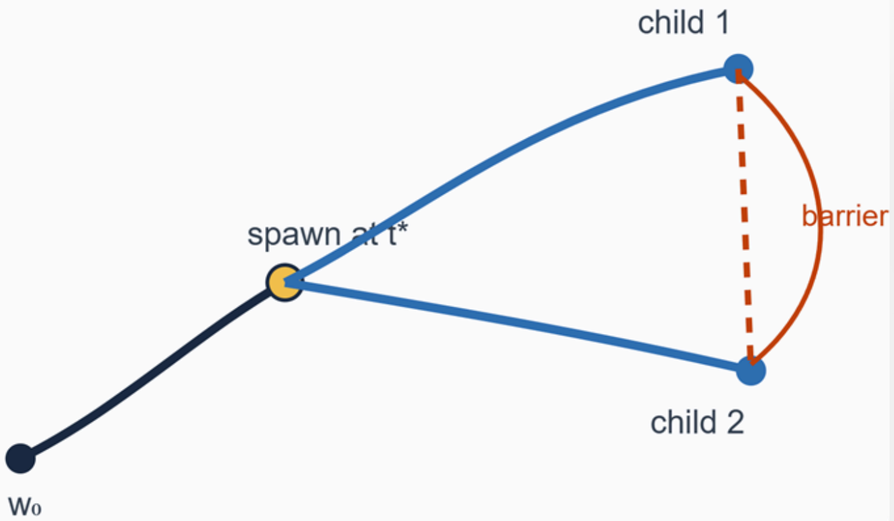
Low error along the whole line means both copies ended in the same valley: the outcome was already fixed at t*. A bump means they ended in different places.

2

Result

On the 1000 lowest-EL2N examples the barrier vanishes very early: easy examples are settled almost at once. On the 1000 highest-EL2N examples it stays high much longer.

3



THE PAPER IN FIVE SENTENCES

1) The influence of a training example on the others is bounded by its gradient norm, which defines GraNd.

2) After a few epochs GraNd is approximated by the error norm EL2N, which needs only a forward pass.

3) Removing low-EL2N examples cuts 50% of CIFAR-10 and 25% of CIFAR-100 without loss in accuracy.

4) The very highest scores are often noisy or atypical, so the best subset skips them.

5) High-score examples drive feature learning and keep the loss landscape rough for longer.



A CRITICAL LOOK

Erratum
Pruning at initialization was a bug. Only scores from trained networks work.

Scoring costs
10 runs × 20 epochs equal one full 200-epoch training. It pays off when the pruned set is reused.

Coverage
Hard-only subsets fail at high pruning rates. Sorscher et al. (2022) show that with scarce data easy examples should be kept.

Scale
All experiments use CIFAR-scale image benchmarks.

DOES IT HOLD UP? OUR REPRODUCTION

ResNet-18 on CIFAR-10, 156 training runs. Details are in the report.

Prune 50% without loss
No loss up to 40%; −0.45 pp at 50%, +2.1 pp over random

MOSTLY YES

GraNd works at initialization
Below random from 20% to 90% pruned

NO

Scores work after a few epochs
Beats random from epoch 2, full effect by epoch 10

YES

Excluding the hardest examples helps
Best at 500 excluded, +0.12 pp, within noise

WEAK

THE DATA DIET PLAN

1. Train a few networks for 10–20 epochs and average EL2N.
2. Prune moderately and skip the very hardest examples.
3. Do not trust scores computed at initialization.

THANK YOU